



# **INDIAN RAILWAYS KAVACH 4.0**

## **REFERENCE GUIDE FOR LOCO PILOTS**

Doc. No. 5 16 75 0056

Version: 1.1

Date: 27.09.2025

Prepared by

Approved by



# INDEX

| <b>S. No.</b> | <b>Item Description</b>                                                                               | <b>Page</b> |
|---------------|-------------------------------------------------------------------------------------------------------|-------------|
| 1             | Loco KAVACH - Subsystems                                                                              | 2           |
| 2             | Loco KAVACH Equipment Layout in Locomotive                                                            | 4           |
| 3             | Driver Machine Interface (DMI)                                                                        | 6           |
| 4             | Loco KAVACH Startup Procedure                                                                         | 8           |
| 5             | Powering Off Loco KAVACH – Diesel/Electric & Conventional, Traction Locomotives / EMUs, Vande Bharath | 9           |
| 6             | Loco KAVACH Configuration Procedure                                                                   | 21          |
| 7             | Mode Transition chart                                                                                 | 23          |
| 8             | Operating Modes: Switching Between Various Modes & Assigned Responsibilities                          | 24          |
| 9             | Mode Transitions and Loco Pilot Responsibilities                                                      | 30          |
| 10            | Procedure for SoS Signal Generation and Reception in Loco KAVACH                                      | 32          |
| 11            | Collision Scenarios                                                                                   | 34          |
| 12            | Do's and Don'ts                                                                                       | 35          |
| 13            | Conclusions                                                                                           | 36          |
| 14            | References                                                                                            | 36          |

## 1.0 Loco KAVACH - Subsystems

The Loco KAVACH unit is the central component of the KAVACH system, ensuring safe train operations through real-time communication and automatic control mechanisms. It works in coordination with several subsystems to enhance safety, automation, and efficiency.

### Loco Vital Computer



- Primary processing unit of KAVACH; executes safety logic and communication.
- Monitors speed, braking, signal aspects, and train positioning.
- Interfaces with BIU, RFID, GPS, and DMI.

### Driver Machine Interface

- Provides real-time alerts, warnings, and operational status.
- Displays signal aspects, braking status, and system faults.
- LP-OCIP (DMI): Interface for loco pilot inputs.



### Brake Interface Unit (BIU)



- Automatically applies brakes if over speeding or signal violation is detected.
- Interfaces with different braking systems:
  - IRAB – Uses HBL BIU.
  - E-70 (Faively Transport) & CCB (Knorr Bremse) – Use OEM BIUs.

### RFID Reader

- Reads track side RFID tags to determine location, direction, and upcoming signal aspects.
- Works in real-time to assist KAVACH in decision-making.



## Speedometer

- Displays current train speed in real-time.
- Integrated with KAVACH to enforce speed restrictions and trigger auto-braking if limits are exceeded.



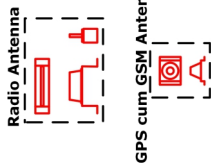
---

## GPS cum GSM Antenna

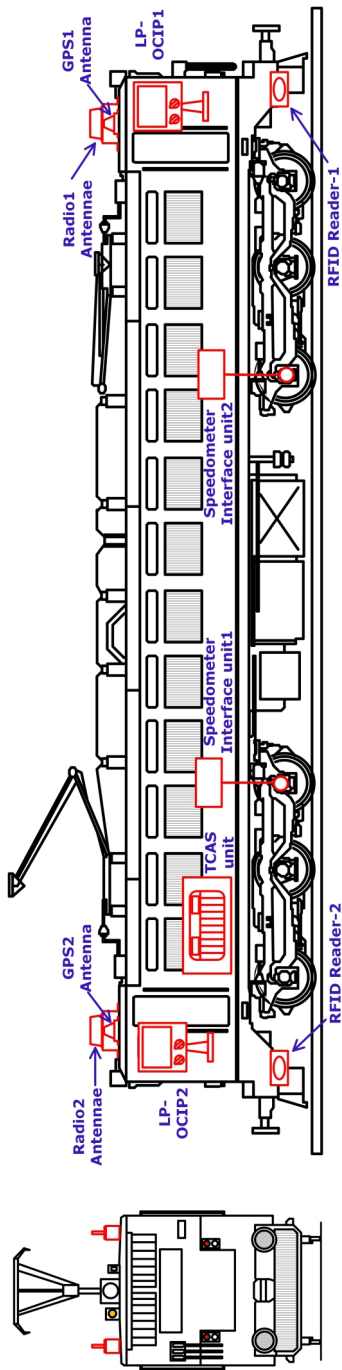


- Provides time synchronization for KAVACH communication.
  - Logs real-time events; dual GPS CUM GSM modems ensure redundancy.
  - Fault detection in 3 cycles, with automatic recovery.
-

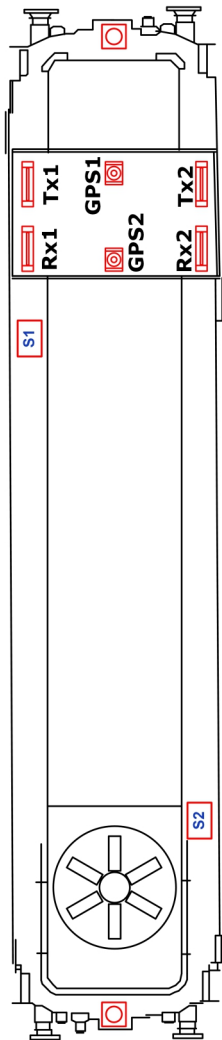
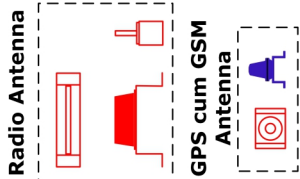
# 3.0 KAVACH EQUIPMENT LAYOUT PLAN FOR ELECTRIC LOCOS



Locomotive Top-view without Pantograph and its circuit

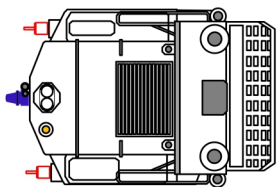


# KAVACH EQUIPMENT LAYOUT PLAN FOR DIESEL LOCOS



Long-Hood side

Short-Hood side



LP-OCIP



RFID Reader



Vital Computer

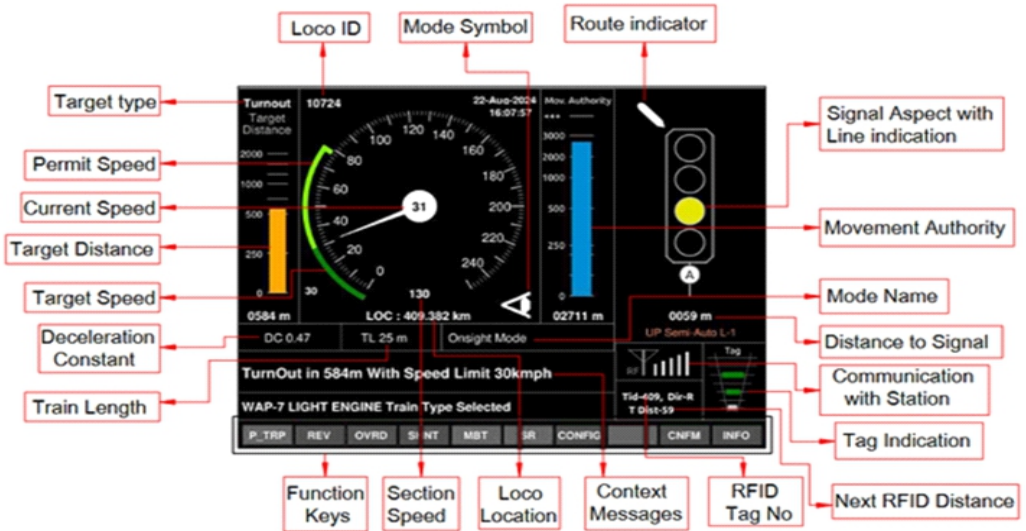


U-BIU



Speedometer

### 3.0 Driver Machine Interface (DMI)



**1. Loco ID:** This field displays the locomotive number.

**2. Current speed:** This field displays the current speed of the locomotive.

**3. Over speed (section speed):** This field turns red if the speed exceeds the permissible limit. The speed indicator needle also turns red, accompanied by an audio buzzer to alert the LP.

**4. Permit speed:** This field displays the permissible speed as regulated by KAVACH in light green. The permissible speed is determined based on loco speed, section speed, Permanent Speed Restrictions (PSRs), turnouts, and SOS conditions.

**5. Target Speed:** This field displays the maximum speed permitted by KAVACH in dark green. The target speed is calculated based on loco speed, section speed, PSRs, turnouts, and SOS conditions.

**6. Target type/ Target Distance:**

**a.Target Type:** This field displays the type of upcoming speed control scenario (e.g., PSRs, turnouts).

**b. Target Distance:** This field displays the distance from the locomotive's current position to the speed control location.

**7.Brake Applied:** This field displays KAVACH-initiated brake application in red. The symbol thickness increases with the brake type:

- Normal Service Brake (NSB)
- Full-Service Brake (FSB)
- Emergency Brake (EB)

**8.Deceleration constant (DC):** This field displays the braking parameters KAVACH uses for applying brakes and controlling deceleration.

**9.Context Message:** This field displays real-time information regarding:

- Approaching turnouts
- Level crossing gates
- Permanent Speed Restrictions (PSRs)
- Failure messages
- Acknowledgment messages

**10.Functional keys:** These keys are used for selecting train configuration and KAVACH operation modes, such as Reverse, Shunt, Isolation, etc.

**11.Train Length:** This field displays the train length computed by KAVACH. The train length is pre-set during initial train configuration and updated after crossing the Advance Starter Signal at each station.

**12.Loco location:** This field displays the current location of the locomotive.

**13.Mode Name/ Symbol:**

- Mode Name: This field displays the current locomotive operation mode in text (e.g., Standby, Staff Responsible, Limited Supervision, Full Supervision).
- Mode Symbol: This field represents the mode using abbreviations (e.g., SHNT for Shunt mode, LS for Limited Supervision).

**14.RF Communication with Station:** This field displays the radio signal strength with stationary KAVACH units at stations, LC, IB, and ABS huts.

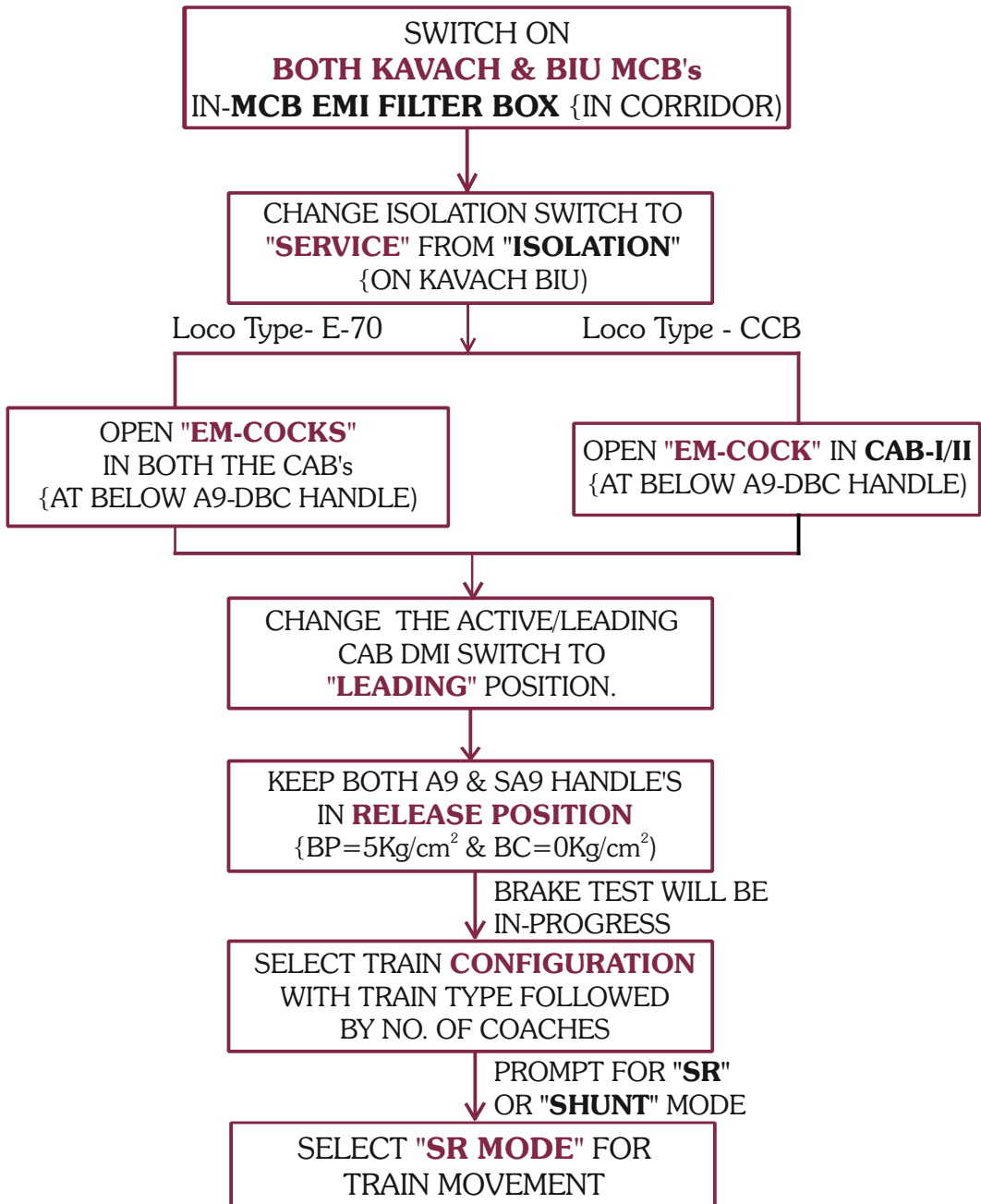
**15. Tag indication:** Tag Indication: This field displays the status of the last three RFID tags read:

- **Green: Successfully read**
- **Red: Not read**

**16. Signal Aspect:** This field displays the current aspect of the approaching signal.

**17. Movement Authority:** This field displays the distance KAVACH permits the train to travel.

# LOCO KAVACH STARTUP PROCEDURE

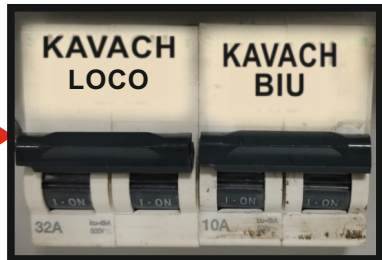


## 5.0) Switch 'ON' Sequence (E70)

1. Switch-ON '**E70-BIU**' MCB first and then '**KAVACH**' MCB provided in MCB EMI FILTER BOX.



**MCBs OFF**



**MCBs ON**

**Figure 1: MCBs to Power-ON KAVACH and E70-BIU**

2. Change the Isolation Switch knob on **E70-BIU (RIB)** unit from '**ISOLATION**' to '**SERVICE**' Position



**ISOLATION Position**



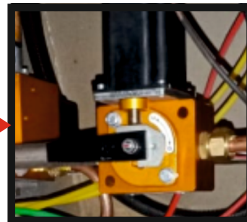
**SERVICE Position**

**Figure 2: E70-BIU (RIB) switch for ISOLATION / SERVICE Selection**

3. Keep the **EM** cock from '**CLOSE**' to '**OPEN**' position under CAB-I/II under driver desk.



**CLOSE-Position**



**OPEN-Position**

**Figure 3: Isolating cocks positions**

4. In active **CAB**, keep Leading / Non-Leading switch on **LP-OCIP (DMI)** in Leading position.



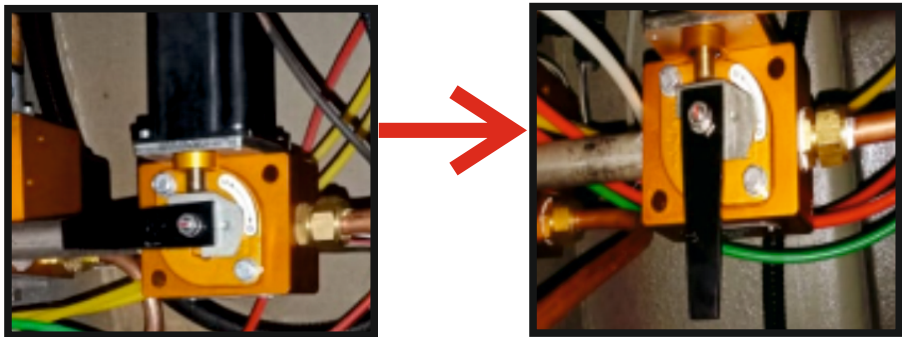
Non Active CAB

Active CAB

Figure 4: Leading/Non-Leading Switch positions

### Switch ‘OFF’ Sequence (E70)

1. Turn both EM cocks from ‘**OPEN-Position**’ to ‘**CLOSE-Position**’ in both CAB.

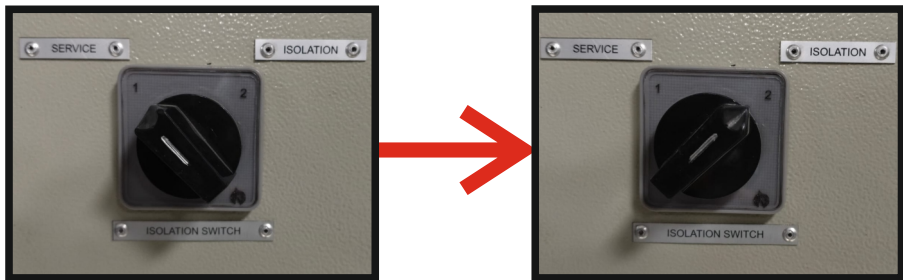


OPEN-Position

CLOSE-Position

Figure 5: Isolating Cocks Position

2. Change the isolation switch on **CCB-BIU(RIB)** from **SERVICE** to ‘**ISOLATION**’ mode.



SERVICE Position

ISOLATION Position

Figure 6

### 3. Switch-OFF the 'KAVACH LOCO' and 'KAVACH E70-BIU' MCBs



Figure 7

### Switch 'ON' Sequence(CCB)

1. Switch-ON 'CCB-BIU' MCB first and then 'KAVACH' MCB provided in MCB EMI FILTER BOX.

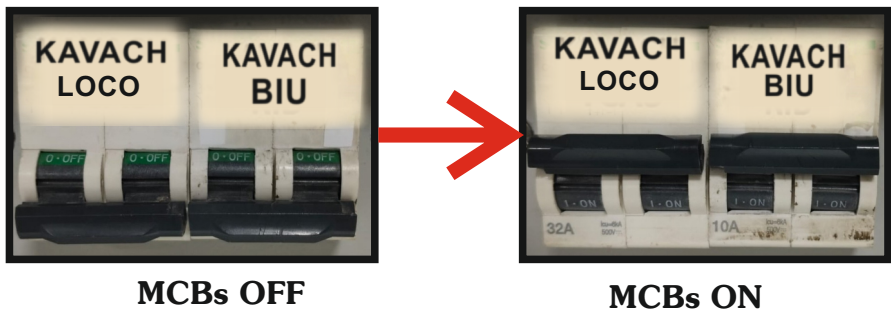


Figure 1: MCBs to Power-ON KAVACH and CCB-BIU

2. Change the Isolation Switch knob on CCB-BIU (RIB) unit from 'ISOLATION' to 'SERVICE' Position

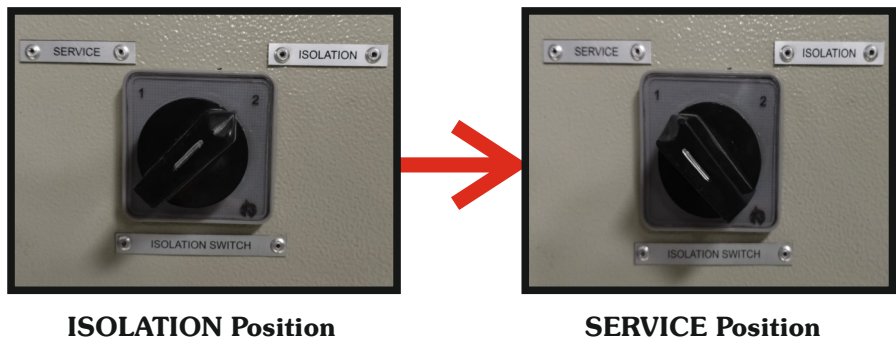
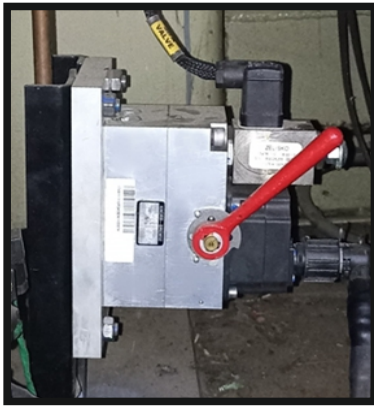
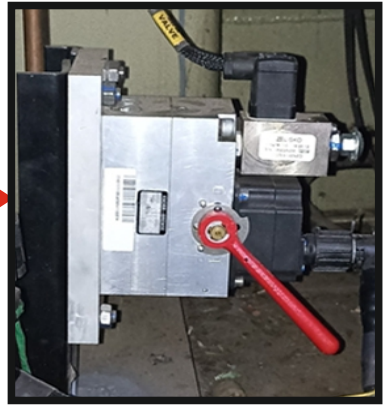


Figure 2: CCB- BIU (RIB) switch for ISOLATION / SERVICE Selection

3. Keep the **EM** cock from 'CLOSE (I)' to 'OPEN(O)' Position' under CAB-I/II under driver desk.



**CLOSE-Position**

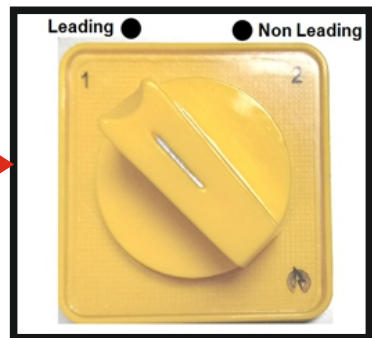


**OPEN-Position**

**Figure 3: Isolating cocks positions**



**Non Active CAB**

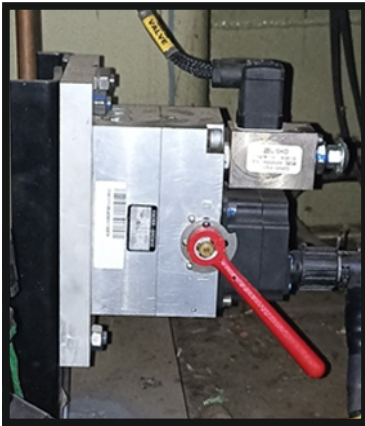


**Active CAB**

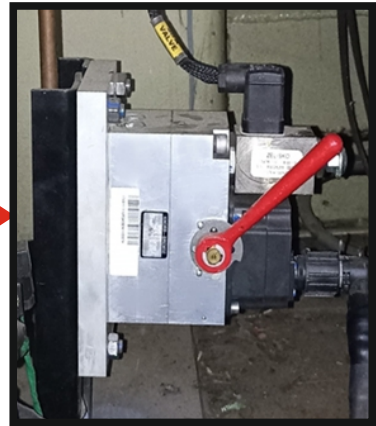
**Figure 4**

## Switch 'OFF' Sequence (CCB)

1. Change EM cock from '**OPEN**' to '**CLOSE**' Position in CAB-I/II



**OPEN-Position**



**CLOSE-Position**

**Figure 5: Isolating Cocks Position**

2. Change the isolation switch on **CCB-BIU(RIB)** from **SERVICE** to '**ISOLATION**' mode.



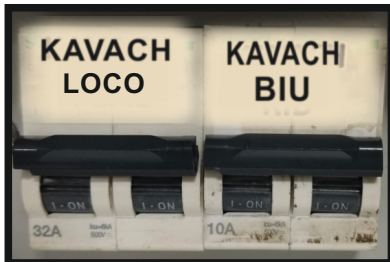
**SERVICE Position**



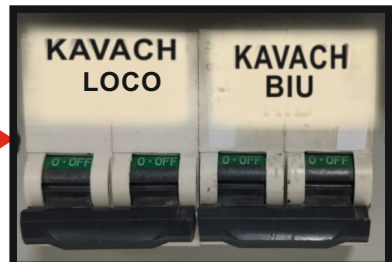
**ISOLATION Position**

**Figure 6**

3. **Switch-OFF** the '**LOCO KAVACH**' and '**CCB-BIU**' MCBs



**MCBs ON**

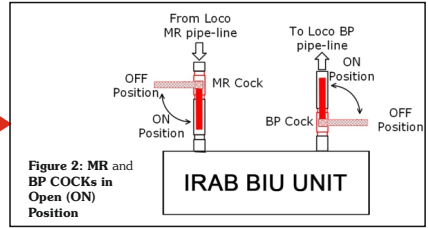
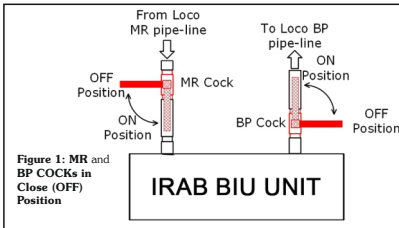


**MCBs OFF**

**Figure 7**

## Switch 'ON' Sequence (IRAB-DIESEL/ELECTRIC)

1. Keep both **MR** and **BP COCKs** from 'CLOSE(OFF)' to 'OPEN(ON)' Position



2. Switch-ON both the 'KAVACH LOCO' and 'KAVACH BIU' Provided in MCB EMI FILTER BOX



**MCBs OFF**



**MCBs ON**

**Figure 3 MCBs to Power ON KAVACH LOCO and IRAB BIU**

3. Change the Isolation Switch knob on **IRAB-BIU (RIB)** unit from 'ISOLATION' to 'SERVICE' Position



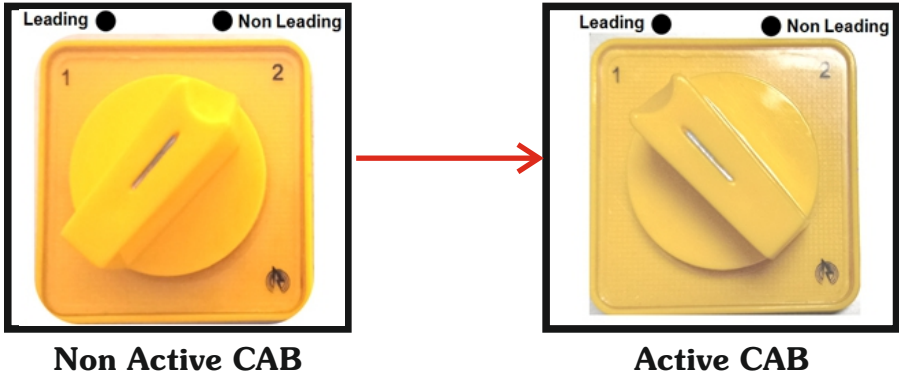
**ISOLATION Position**



**SERVICE Position**

**Figure 4: IRAB-BIU (RIB) Isolation switch for ISOLATION / SERVICE Selection**

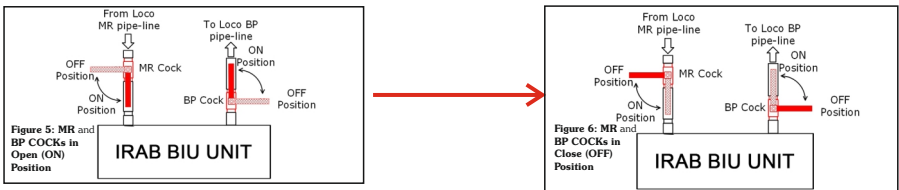
4. In active **CAB**, keep Leading / Non-Leading switch on **LP-OCIP (DMI)** in Leading position.



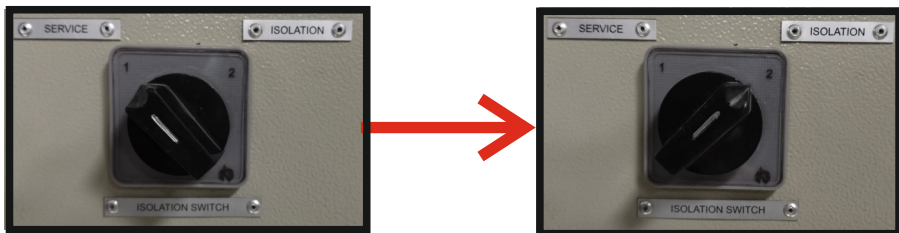
**Figure 4: Leading/Non-Leading Switch positions**

## Switch ‘OFF’ Sequence (IRAB-DIESEL/ELECTRIC)

1. Keep both **MR** and **BP COCKs** from 'OPEN(ON)' to 'CLOSE(OFF)' position



2. Change the isolation switch on **IRAB-BIU(RIB)** from **SERVICE** to 'ISOLATION' mode.

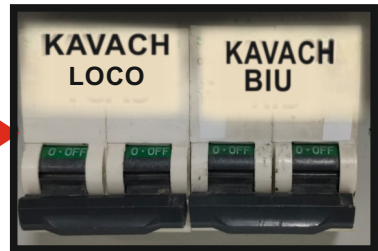


**Figure 6 IRAB-BIU (RIB) Isolation switch for SERVICE /ISOLATION Selection**

### 3. Switch-**OFF** both the '**KAVACH LOCO**' and '**KAVACH BIU**' provided in **MCB EMI FILTER BOX**



**MCBs ON**



**MCBs OFF**

**Figure 7 MCBs to Power OFF KAVACH LOCO and IRAB BIU**

## **DMR MCB Switching for Diesel Locomotive**

1. DMR MCB for DMR circuit to be kept in '**OFF**' position.



PCF  
DMR  
MCB

**Figure 1**

2. If Loco Pilot experiences no throttle, Switch '**ON**' the DMR MCB

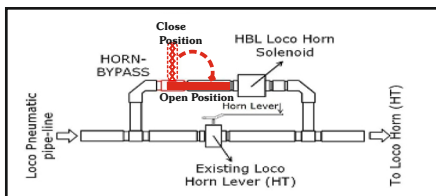


PCF  
DMR  
MCB

**Figure 2**

## **Loco Horn isolation in LOCO KAVACH(E70/CCB/IRAB)**

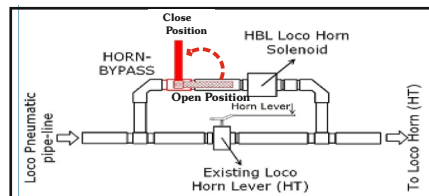
1. Change the auto Horn (Bypass) from '**Close**' to '**Open**' position for auto horn enable.



**Figure 1**

Horn Bypass cock '**OPEN**'(ON) Position

2. In case of continuous Loco horn due to any failure on kavach side, move the **HORN-BYPASS** cock to '**CLOSE**'(OFF) Position'.

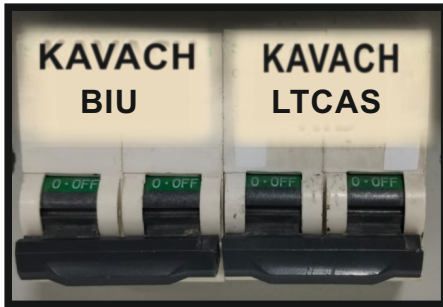


**Figure 2**

Horn Bypass cock '**CLOSE**'(OFF) Position

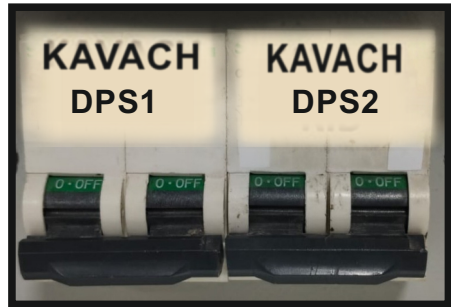
# Switch 'ON' Sequence (VANDE BHARATH)

1. Switch-ON First **KAVACH & BIU** Followed by **DPS1& DPS2**.

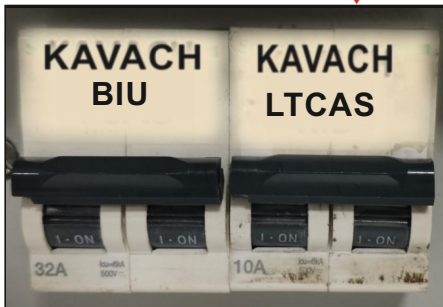


**MCBs OFF**

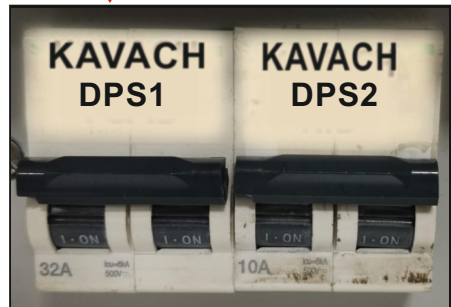
**Figure 1: MCBs to Power OFF/ON KAVACH DPS1, DPS2, BIU and LTCAS**



**MCBs OFF**



**MCBs ON**



**MCBs ON**

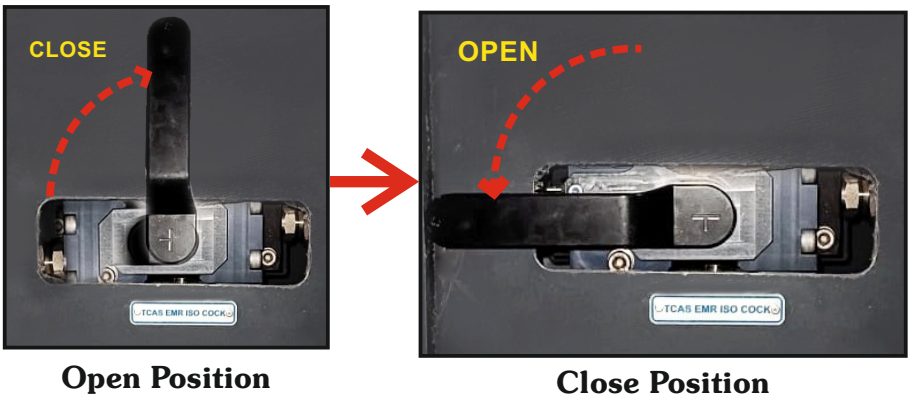
**Figure 2: MCBs to Power OFF/ON KAVACH DPS1, DPS2, RIB and LTCAS**

2. Change the Isolation Switch Knob on **VB KAVACH PANEL(BIU)** unit from '**ISOLATION**' to '**SERVICE**' Position



**Figure 2: KAVACH PANEL(BIU) Isolation switch for ISOLATION / SERVICE Selection**

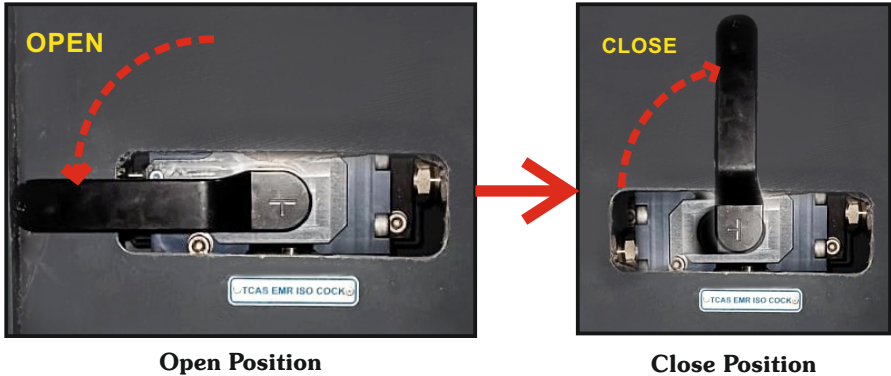
3. In active CAB, keep KAVACH EM COCK from '**CLOSE** Position to '**OPEN** Position'.



**Figure 3 ISOLATING COCK Position**

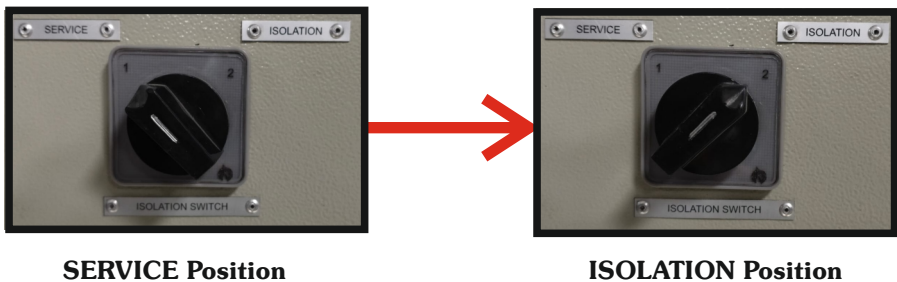
## Switch 'OFF' Sequence (VANDE BHARATH)

1. In active CAB, keep **KAVACH EM COCK** from 'OPEN' Position to 'CLOSE' Position'.



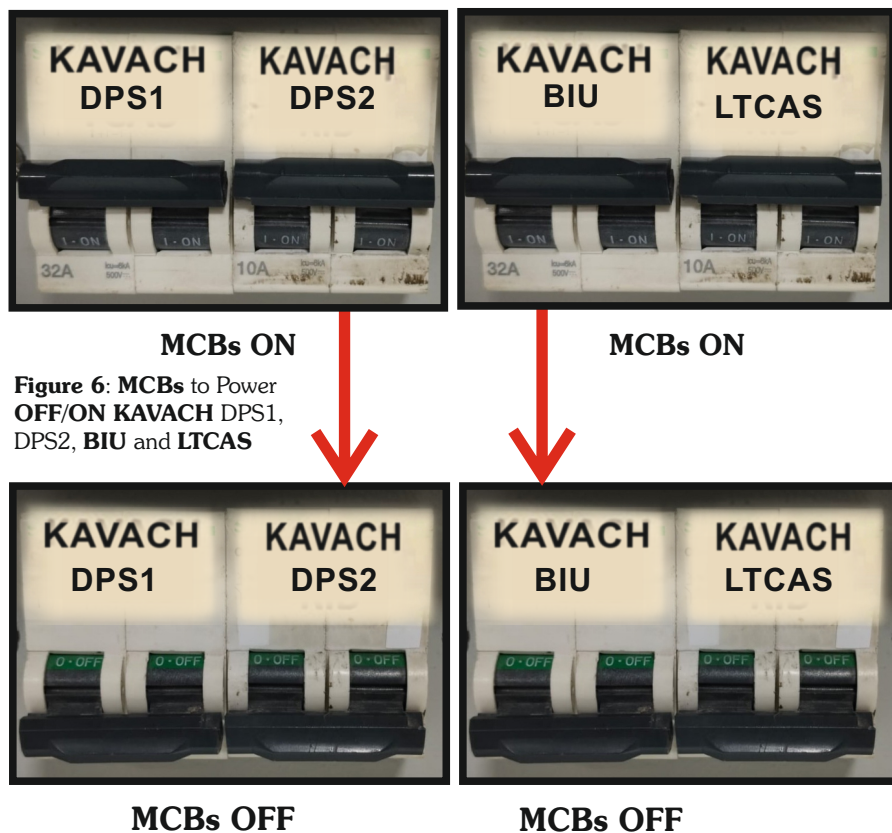
**Figure 4 ISOLATING COCK Position**

2. Change the Isolation Switch Knob on **VB KAVACH PANEL(BIU)** unit from 'SERVICE' to 'ISOLATION' Position



**Figure 5: KAVACH PANEL(BIU) Isolation switch for SERVICE/ISOLATION Selection**

3. Switch-OFF the 'KAVACH DPS1, DPS2 and 'KAVACH LOCO, KAVACH BIU' provided in KAVACH Panel



**Figure 7: MCBs to Power OFF/ON KAVACH DPS1, DPS2, BIU and LTCAS**

## 4. The following Kavach startup procedure must be executed at Start of Mission.

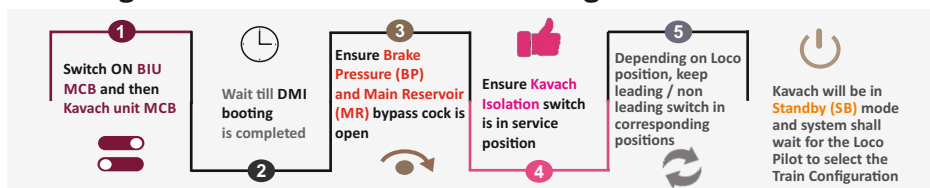
### 4.1. For E-70 Loco's:

- Turn ON both the **KAVACH & BIU MCB's** located in SB1 panel / **MCB EMI** Filter Box
- On the BIU unit, keep the BIU switch in SERVICE position.
- Open the EM Cocks (located below A9 handle) in both CAB-I & CAB-II.
- Ensure that IRU switch is in Normal or Service position
- In the active CAB, set the KAVACH DMI switch to LEADING position.
- KAVACH performs A9 brake tests by dropping BP in 3 stages:
  - NSB (Normal Brake): BP drops by  $0.5 \text{ kg/cm}^2$  ( $\sim 4.6 \text{ kg/cm}^2$ ).
  - FSB (Full Service Brake): BP drops by  $1.5 \text{ kg/cm}^2$  ( $\sim 3.5 \text{ kg/cm}^2$ ).
  - EB (Emergency Brake): BP drops to  $0 \text{ kg/cm}^2$ .

### Note:

-  In some locos, after NSB is applied, BP may stop at  $4.7 \text{ kg/cm}^2$  due to BP over charge ( $> 5.3 \text{ Kg/cm}^2$ ) in Loco. In such a case, Kavach DMI displays "waiting for BP  $4.7 \text{ kg/cm}^2$ ".
  - For KAVACH brake test to proceed, move the A9 handle until BP drops to  $4.6 \text{ kg/cm}^2$
- Now, KAVACH will perform SA9 brake test for Loco Brake Cylinder (BC) pressure control:
    -  For this test, SA9 handle must be kept in RELEASE position. Brake test cannot be successfully completed, if SA9 handle is not in RELEASE position, and DMI will display "Waiting for BC..."
    - LE brakes will be applied by KAVACH and BC will rise from  $0 \rightarrow 3.5 \text{ kg/cm}^2$
  - If the above tests are successful, "Brake Testing Success" message appears on KAVACH LP-OCIP.

### Powering on Loco KAVACH - Initial Energization



#### 4.2. For CCB locos:

- a) Turn ON both the KAVACH & RIB MCB's located in SB1 panel / **MCB EMI** Filter Box.
- b) On the BIU unit, keep the BIU switch in SERVICE position.
- c) Open the EM(SIFA) Cock in CAB-II, located below A9 handle. In some locos, this cock may be located in CAB-I.
- d) KAVACH performs A9 brake tests by dropping BP in 3 stages:  
NSB (Normal Brake): BP drops by  $0.5 \text{ kg/cm}^2$  ( $\sim 4.6 \text{ kg/cm}^2$ ).  
FSB (Full Service Brake): BP drops by  $1.5 \text{ kg/cm}^2$  ( $\sim 3.5 \text{ kg/cm}^2$ ).  
EB (Emergency Brake): BP drops to  $0 \text{ kg/cm}^2$ .  
 After EB is applied by KAVACH, A9 handle must be kept in EMER position
  - After "OK TO RUN" message appears on EBV display, A9 handle must be shifted to RUN position for restoring BP to  $5.0 \text{ kg/cm}^2$ .
- e) After A9/BP drop tests, KAVACH performs SA9 brake test for Loco Brake Cylinder (BC) pressure control:  
 For this test SA9 handle must be kept in RELEASE position.
  - LE brakes will be applied by KAVACH and BC will rise from  $0 \rightarrow 3.5 \text{ Kg/cm}^2$ .
- f) If the above tests are successful, "Brake Testing Success" message appears on KAVACH LP-OCIP.

#### 4.3 Configuration of Train:

- a) After the above tests, the system will ask for Train configuration and displays "Select Train configuration" on LP-OCIP.
- b) Press CONFIG & CNFM keys. Select the Train type and number of coaches/wagons from the train configuration menu using the soft arrow keys.
- c) The system will ask for mode selection, displaying "Select SR or shunt mode" on LP-OCIP.
- d) Press SR or SHNT, then CNFM to confirm.
- e) Both Train type and mode shall display on LP-OCIP.

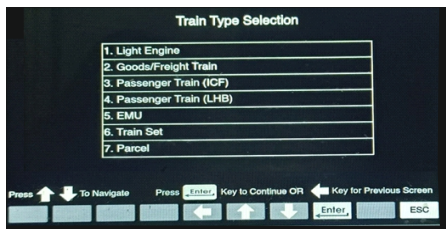
# LOCO KAVACH CONFIGURATION PROCEDURE



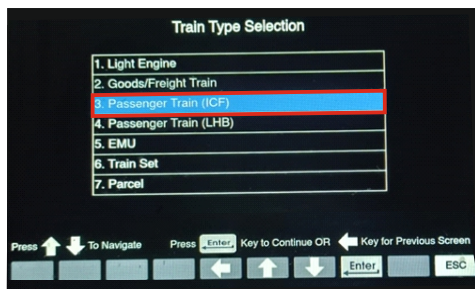
**Figure :1 Press CONFIG & CNFM Buttons**

01. Select 'Loco Configuration' as given below:

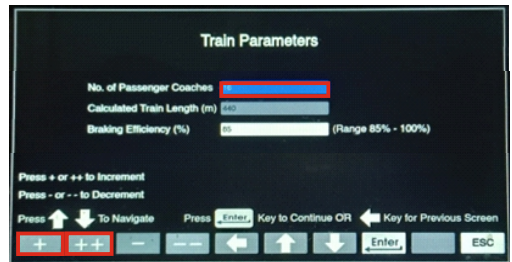
- Press '**CONFIG**' key first and then press '**CNFM**' key on the **LP-OCIP** (DMI) to enter into loco configuration mode
- A menu shall appear on the screen showing loco configurations Light Engine, Goods / Freight Train and Passenger Train
- Press '**UP**' or '**DOWN**' keys to select the type of loco configuration and press '**ENTER**' key to confirm the selected configuration



**Figure 2: Loco Configuration menus on LP-OCIP (DMI) Display**

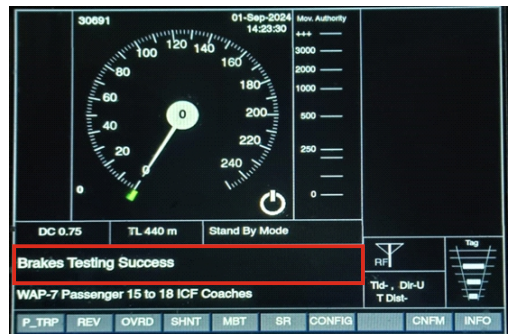


**Figure 3: Selection of Passenger Train Type(ICF / LHB)**

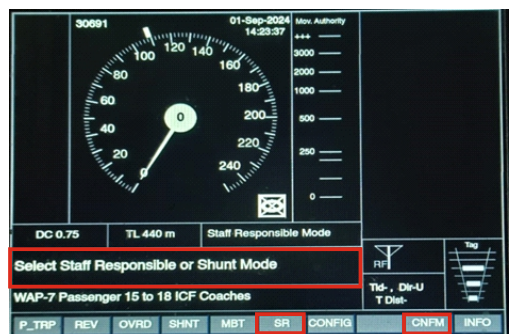


**Figure 4**

- 1.Enter the number of Passenger Coaches (Press soft key “++” for “5” coaches and “+” for “1” coach) and press “Enter” key.
- 2.Passenger coaches selection to be done again.



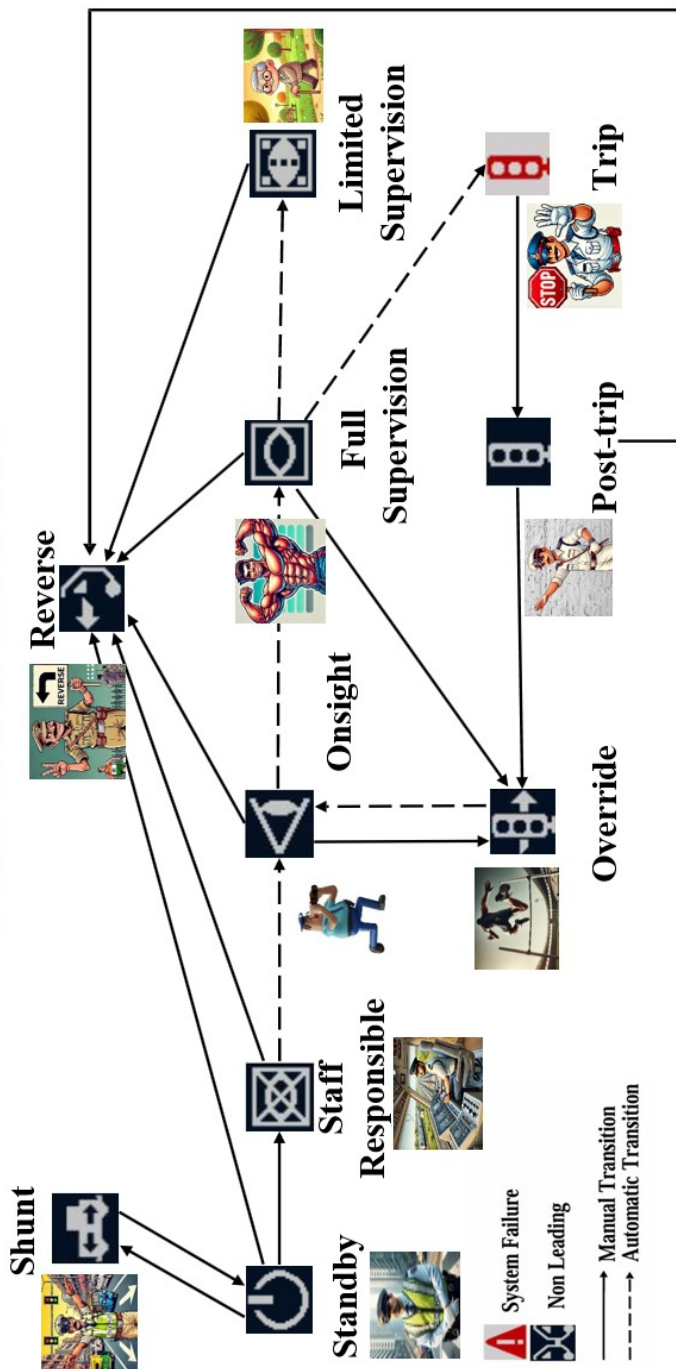
**Figure :5 Brake Testing Success** Message is displayed.



**Figure :6**

Select **SR** and **CNFM** buttons, Loco KAVACH will enters into “Staff Responsible Mode”.

## 7.0) Mode Transitions in Ver.4.0



## 8.0) Operating Modes

### A) Standby (SB) Mode



- This is the default mode and Loco Pilot cannot select it
- The system enters Stand By mode when :
  - It turns ON and completes self tests
  - No CAB is occupied or EM cock is closed In this mode,
  - The Loco pilot can only interact when the desk is open
  - The system monitors train standstill
  - It shows train speed as zero and permitted speed as zero
  - If any movement is found, the Emergency Brake (EB) is applied



### B) Shunt (SHNT) Mode



- This mode can only be selected manually by Loco pilot when the train is at standstill
- It can be selected from SB,SR,LS or FS mode by pressing SHNT,then CNFM
- Onboard Kavach supervises the shunting speed(default:15kmph)
- Kavach does not monitor shunt signals, so it cannot supervise passing them at danger
- The Loco KAVACH unit supervises the train with a shunting speed limit of 15 kmph (default) within shunting limits
- If the train exceeds the Block Section Limit,Stationary KAVACH generates an SOS signal



### C) Staff Responsible (SR) Mode



- This mode can be selected manually by Loco pilot when the train is at standstill
- It can be selected by pressing SR, then CNFM only at the start of mission
- In this mode, the Loco Pilot operates the train under his own responsibility within the KAVACH territory
- It lets the train send and receive emergency stop signals as long as train's ID, direction and location are known.
- The system enters Staff Responsible mode when :
  - o Loss of radio communication in communication mandatory zone
  - o Three consecutive RFID tags are missed
  - o Train direction not known
  - o Kavach Territory exit tag is read



### D) On Sight (OS) Mode



- At the start of mission, Kavach sets the train's direction using RFID tags and then switches to OS mode
- On a Calling ON route in FS mode, Kavach automatically enters OS mode, letting the Loco pilot monitor track conditions
- This mode allows entry into an occupied track section
- Kavach limits speed to 30kmph for Home and Starter signals and Maximum Permissible Speed for LSS & IBS signals
- This mode exits when train passes the next stop signal at OFF aspect



## E) Full Supervision (FS) Mode



- This mode activates automatically when all Kavach conditions are met and cannot be selected by the Loco Pilot
- This mode starts when train and track data, including speed limits or movement authority are available
- The system tracks train movements using a dynamic speed profile
- It provides train speed, permitted speed, target distance and target speed for safe operation



## F) Limited Supervision (LS) Mode



- This mode is not selected by the Loco pilot
- LS Mode activates automatically if radio communication fails for 30 seconds in absolute section and 15 seconds in auto block section, as long as the track data is available up to 3000m
- This mode ensures speed supervision based on the track layout
- LS mode exits when communication is restored, On Sight Movement Authority is received or the loco pilot selects Reverse or Shunt Mode
- The loco pilot must manually follow all signals and speed limits in this mode



## G) Override (OVRD) Selection



- Loco Pilot selects Override by pressing OVRD, followed by CNFM
- Override is permitted only when Movement Authority (MA) is below 200 meters and the train is at standstill
- For LSS/IB signals, Override is allowed even when train is moving at permitted section speed.
- If OSMA is extended or override times out (default: 240 seconds) and the loco pilot does not start movement, the system switches to previous mode
- Kavach monitors override speed of :
  - o 30 kmph for Home & Starter signals and
  - o Permitted section speed for LSS,IB and Gate signals
- The system also obeys SOS signals from the Stationary Kavach
- Trip/Override counter reading must be checked and recorded



## H) Trip Mode



- Trip mode activates when train passes a stop signal at ON or exceeds End of Authority by 30m
- Stationary KAVACH sends an SOS signal, causing the Onboard Kavach to apply continuous braking
- In this mode, the system ensures the train remains stationary through Standstill Supervision
- To release brakes, the loco pilot must press P\_TRIP, followed by CNFM
- Trip/Override counter reading must be checked and recorded



## I) Post Trip Mode

- Loco Pilot activates Post Trip Mode by pressing P\_TRIP, followed by CNFM which releases the Brakes
- To exit Post Trip Mode, loco pilot must select OVRD, followed by CNFM.
- Kavach shall switch to Onsignal mode
- Loco Kavach unit supervises the train with a ceiling speed of :
  - o 30 kmph for Home & Starter signals
  - o Section speed for LSS or IB and
  - o 15 kmph in day time and 8 kmph at night for automatic signals



## J) Non-Leading Mode

- Loco Pilot must manually select this mode when the train is at standstill
- This mode is used for the slave engine whether it is electrically coupled to the leading engine (remote-controlled) or not coupled (non-remote-controlled)
- In Non-Leading Mode, the Loco KAVACH does not monitor train movement



## K) Reverse Mode

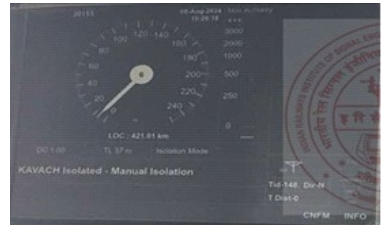
- Loco pilot activates REV mode by pressing REV, then CNFM
- This mode allows the Loco Pilot to change the train's direction while staying in the same cab, without changing the train's orientation
- By default, Kavach limits :
  - o Ceiling speed of 25 kmph
  - o Maximum reverse distance of 500m and
  - o Duration to 300 seconds
- Signal aspects, MA and RFID tag information is not necessary in this mode.



## L) Isolation Mode



- This mode is used by the Loco Pilot when Kavach malfunctions
- In this mode, Onboard Kavach is disconnected from the brake system and does not supervise train movement
- The DMI will clearly indicate that Onboard Kavach is in Manual Isolation
- Isolation counter reading must be checked and logged with a reason



## M) System Failure (SF) Mode



- Activated when a fault affecting safety is detected.
- Onboard KAVACH commands the Emergency Brakes.
- The Loco Pilot must isolate the KAVACH system and operate the train under their responsibility.
- Exits when the Loco Pilot isolates the system and takes control.



## 6.SOS Generation and Reception



### a) Manual SOS

- Loco Pilot presses SOS and COMMON on the DMI to send an emergency alert
- This alert notifies nearby trains and stations about the emergency

### b) Unusual Stoppage SOS

- If the train stops unexpectedly despite having a clear track or when MA exceeds 300m, Kavach prompts the loco pilot with buzzer
- Loco pilot must press Common/ACK on DMI within 15 seconds
- If not acknowledged, Kavach sends an SOS and applies brakes to nearby trains

### c) SOS Reception & Recovery

- When Onboard Kavach receives an SOS signal from another Loco or Station, nearby trains apply brakes and stop
- After stopping, trains are limited to 30kmph
- Normal speed is restored when SOS is cancelled or the train moves away
- The SOS counter reading must be checked and logged

## **9.0. Mode Transition and Responsibility of Loco Pilot**

### **9.1. Mode Transition:**

#### **a) Standby Mode to Shunt Mode**

After train configuration, Loco KAVACH prompts the Loco Pilot to select either Staff Responsible (SR) Mode or Shunt Mode. If Shunt Mode is selected, the system transitions into Shunt Mode, provided the train is at a standstill in Standby Mode.

- In Shunt Mode, Onboard KAVACH supervises speed limits (default: 15 kmph) within designated shunting limits.
- The Loco Pilot is responsible for adhering to shunting limits and ensuring train movements comply with Station Working Rules and General & Subsidiary Rules (G&SR).

#### **b) Standby Mode to Staff Responsible (SR) Mode**

After train configuration, Loco KAVACH prompts for the selection of SR Mode or Shunt Mode. If SR Mode is selected, the system transitions into SR Mode from Standby Mode.

#### **c) Staff Responsible Mode to On-Sight Mode**

- This transition occurs automatically when:
- Loco KAVACH detects two RFID tags sequentially.
- RF Communication is established with the Station or LC KAVACH.

#### **d) Full Supervision / On-Sight Mode to Trip Mode**

Loco KAVACH enters Trip Mode under the following conditions:

1. Train passes a Stop Signal at ON or reaches the End of Authority (EOA).
2. Onboard KAVACH malfunctions, leading to system safety intervention.
3. Signal reverts to danger (ON) unexpectedly in the train's approach, without adequate braking distance.

#### **e) Full Supervision to Limited Supervision Mode**

Limited Supervision Mode cannot be selected manually by the Loco Pilot. The system automatically transitions from Full Supervision to Limited Supervision Mode when:

- Valid radio packets are unavailable for a configured duration (default: 30 seconds) within a Communication Mandatory Area.
- Track profile information is available for at least 3000 meters (configurable).

## **f) On-Sight Mode to Full Supervision Mode**

Full Supervision Mode cannot be selected manually by the Loco Pilot. It is automatically activated when:

- The train, in On-Sight Mode, has crossed at least one Stop Signal in OFF (proceed) condition.
- Track profile is available (including Traffic Direction) for 3000 meters or up to the current Movement Authority (MA), whichever is lesser.
- Valid radio packets are being received within the Communication Mandatory Area.
- The Movement Authority (MA) remains valid.
- g) Full Supervision Mode to Staff Responsible (SR) Mode
- The system automatically switches from Full Supervision to SR Mode when:
- The train fails to detect three consecutive RFID tags along the track.

## **9.2. Responsibility of Loco Pilot**

Train movement from KAVACH section to non-KAVACH fitted section:

KAVACH will switch to SR mode and Loco pilot will be prompted to acknowledge after exit from KAVACH territory in within 15 sec.

- If he fails to acknowledge, brakes will be applied by the Loco KAVACH.

## **10.0 Procedure for SoS Signal Generation and Reception in Loco KAVACH**

1. As all the locomotives in the section may not be fitted with KAVACH, procedure laid down in G & SR to be followed.

2. When a train comes to a stop between stations or at a station, on account of any accident or any cause which is not immediately obvious (including tripping of traction power on OHE in electrical section) and the Loco Pilot finds that his train cannot proceed, he shall immediately protect the train as per provisions of GR 6.03.

- The train fails to detect three consecutive RFID tags along the track.

### **Generation of manual SoS**

When a loco pilot detects an emergency and needs to alert other KAVACH-equipped trains in the territory, they must press the 'SoS' and 'COMMON' push buttons on the DMI simultaneously to generate a manual SoS from the Active CAB. It's important that KAVACH-equipped trains are in the 'on' condition.

### **Unusual Stoppage in block section (SoS)**

If train stops in block section in spite of Movement authority, KAVACH asks LP for acknowledgement of Block section Stop. He has to press 'Acknowledge/Cancel' button on DMI within 15 seconds. The below mentioned message will be displayed on DMI/OCIP.

### **Ack Block Section Stop, SoS Generates in 10s**

If not acknowledged, loco KAVACH broadcasts Save our Soul (SoS) message and brakes will be applied. the below mentioned message will be displayed on DMI/OCIP.

### **SoS-Self loco (stopped in Block)**

- The Loco/Train approaching towards the originating Location of "Side Collision" message, on getting "Side Collision/Unusual Stoppage."
- Message shall display it on the DMI & shall automatically apply brakes to bring the train/locomotive to standstill before reaching the originating Location of "Side Collision".
- Message. After the speed is reduced to Zero kmph, train speed shall be supervised for 30Kmph (configurable) till the train passes the originating Location of 'Side Collision' message.
- The Loco/Train approaching towards the originating Location of "Side Collision" message, on getting "Side Collision/Unusual Stoppage."

- Message shall display it on the DMI & shall automatically apply brakes to bring the train/locomotive to standstill before reaching the originating Location of “Side Collision”.
- Message. After the speed is reduced to Zero kmph, train speed shall be supervised for 30Kmph (configurable) till the train passes the originating Location of ‘Side Collision’ message.

### **Save Our Soul (SoS) Reception consequences and normalization:**

When Loco KAVACH receives an SoS from any loco KAVACH/Station KAVACH in the ON condition, trains approaching the originating source of the SoS (within 3000m of the location of the SoS originating station) shall apply brakes and come to a stop. Following this, a speed supervision of 30 kmph will be enforced until the Train passes the source of SoS generation. Normal Speed will be restored only under the following conditions:

- SoS message is Cancelled by the source.
- Train/Loco is moved away to more than 1500m from the SoS source.
- SoS message is not received from the source for more than 3 minutes.
- The below mentioned message will be displayed on DMI/OCIP when SoS is received.

### **SoS – From station SoS to this Loco (512)**

Note: Station and locomotives have been provided with unique ID for identification. Whenever the save our sole (SoS) is generated Driver Machine Interface (DMI) will display Loco/Station ID.

## 11.0 Collision Scenarios

**Head on collision:** whenever two KAVACH equipped trains enter into same line in Block section and starts to move in opposite direction, Loco KAVACH detects this situation (Only when radio communication is available between two KAVACH equipped Trains) and alerts the loco pilot to apply brakes to avoid collision. If Loco Pilot really wants to move ahead, LP can select shunt mode/non leading mode.

Message as shown below is displayed on DMI / OCIP on detection of head on collision scenario.

**“Head on collision with loco 233318 in 4500 m”**

**Rear end collision:** When two KAVACH equipped trains enter same line in block section and starts moving in same direction, rear train Loco KAVACH detects this situation (only when radio communication is available between two KAVACH equipped trains), and alerts the loco pilot to apply brakes to avoid collision. If Loco Pilot really wants ahead, can select Shunt mode. Leading Loco KAVACH have no impact in this situation.

Message as shown below is displayed on DMI/OCIP on detection of rear end collision scenario.

**“Rear End Collision with Loco 235641 in 5400 m”**

## 12.0 Do's and Don'ts

### DO'S

|    |                                                                                                                                                                                                                                                                                                         |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Verify Loco KAVACH, DMI equipment & BIU before switching on the Onboard KAVACH. If any abnormality Loco KAVACH is to be treated as defective. LP shall intimate / inform respective loco shed and concerned Traction Loco Controller ITLCJ / Power Controller. (PRCJ that the Loco KAVACH is defective) |
| 2  | Read and understand this User Manual thoroughly along with OEMs issued user manuals.                                                                                                                                                                                                                    |
| 3  | Acknowledge the alerts on DMI, immediately within the specified time.                                                                                                                                                                                                                                   |
| 4  | Properly select the train configuration (Goods/ Formation/ Light Engine) and operating modes.                                                                                                                                                                                                           |
| 5  | Open MR and BP valves, Put BIU switch in service mode and switch ON required MCBs when activating Loco KAVACH.                                                                                                                                                                                          |
| 6  | Close MR and BP valves, Put BIU switch in Isolation mode and switch off all MCBs when switching off Loco KAVACH.                                                                                                                                                                                        |
| 7  | Use the DMI with care.                                                                                                                                                                                                                                                                                  |
| 8  | Loco pilot has to inform Traction Loco Controller (TLC) / Power Controller (PRC) when Loco KAVACH is defective, so as to attend to the failure by concerned Railway/OEM staff at the earliest.                                                                                                          |
| 9  | LP shall follow the G & SR provisions and operational requirements issued by Railways.                                                                                                                                                                                                                  |
| 10 | The counter reading of Loco KAVACH equipment will be recorded in Maintenance Register/Sheet and analyzed by concerned Railway Staff and OEMs during the visit of Loco to the shed.                                                                                                                      |

### DON'Ts

|   |                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------|
| 1 | Switch Onboard KAVACH and DMI are found broken.                                                         |
| 2 | Attempt any repair work                                                                                 |
| 3 | Switch OFF power to the Loco KAVACH system, when it is functioning.                                     |
| 4 | Remove system modules or connectors, Radio Modems and Antennae connections when KAVACH is in operation. |
| 5 | Isolate BIU switch when Loco is in movement.                                                            |

## **12.0) Conclusions**

1. KAVACH is a safety aid to loco pilot and would intervene only if LP is not able to react to the unsafe situations.
2. Loco pilot must follow line side signals, caution orders and whistle as per the prevalent operating rules. The operation of KAVACH shall, no way, infringe/override the rules of normal train operations on zonal railway as per G&SR.
3. In case of any clarifications required please refer RDSO specifications, Guidelines issued on time to time on KAVACH Ver 4.0 ,KAVACH manuals approved by RDSO.

## **13.0) References**

1. RDSO specification of KAVACH No. RDSO/SPN/196/2020/ Version 4.0.
2. FRS of KAVACH (IRATP System), Specification No. RDSO/SPN/196/2020 version 4.0

**HBL<sup>®</sup>**

**HBL ENGINEERING LIMITED**

Road No. 10 Banjara Hills, Hyderabad. 500 034 TG, India | [www.hbl.in](http://www.hbl.in), E-mail: [contact@hbl.in](mailto:contact@hbl.in)

**HBL HELPDESK: 18005991799**